

Matthew Oj

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SKILLS

Languages: Python, SQL, JavaScript/TypeScript, PHP, HTML, CSS, MATLAB, Bash

Frameworks & Cloud: FastAPI, Next.js, React, Astro, Tailwind, Laravel, Docker, GitHub Actions, AWS, GCP, Vercel, Netlify, CI/CD

EXPERIENCE

Vafo Digital

London, UK

Senior Data Engineer (Backend Infrastructure) | Python, SQL, AWS, Docker, Snowflake

Apr 2026 – present

- Led migration of fragmented backend orchestration (Snowflake procedures, PHP services, AWS lambda functions) to a unified self-hosted platform on AWS EC2 using Docker, provisioning isolated staging/production environments with an Aurora RDS metadata store and IAM-scoped access, consolidating scheduling for 100+ services
- Architected a reporting service (FastAPI, AWS Lambda, Mangum) automating scheduled delivery across 10 client accounts, eliminating manual ad hoc requests
- Built a shared Python library packaged for reuse, standardising error handling and logging across 30+ services and 4 integrations (Klaviyo, DynamoDB, SFTP, S3)
- Architected a Snowflake Cortex AI system analysing Snowplow tracking and customer behaviour data to generate personalized ad creatives in real time, implementing role-based access controls and secure MCP integration (ongoing, 1 client)

Data Engineer | PHP, Laravel, Snowflake, JavaScript, HTML, CSS

Apr 2025 – Apr 2026

- Collaborated with the Data Science Team to productionise 4 machine learning pipelines using dbt feature engineering, XGBoost/SHAP training, and Snowflake Model Registry for versioning
- Built and hosted 3 client-facing analytics platforms (Python, Streamlit) on Snowflake, with high-performance querying against event tables exceeding 200M+ rows to deliver real-time insights to non-technical stakeholders
- Developed Python AWS Lambda services syncing backend data to external systems (HubSpot, Klaviyo), supporting 250K+ users

Maison Privé Concierge Limited

Remote

Software Engineer (Freelance) | Typescript, Astro, Tailwind, Airtable, GCP, Vercel | [Link](#)

Apr 2026 – Jun 2026

- Architected and built a luxury concierge marketing site with another developer, designing backend integrations (Mailgun, Google Drive, Airtable) to route enquiries through multiple services with retry logic and idempotency checks ensuring data consistency.
- Engineered a dynamic enquiry form with per-service field schemas and strict datatype validation, implementing multi-layer spam filtering that blocked 95%+ of bot submissions while maintaining low friction for genuine customers.
- Deployed on Vercel using TypeScript, Astro, and Tailwind CSS learning the stack on the job with all backend integrations handling incomplete writes and service failures transparently to the end user.

EssenceMediacom

London, UK

Data Software Engineer | Python, SQL, dbt, GCP

Nov 2023 – Apr 2025

- Built a Cloud Run-hosted web app for table schema generation, cutting pipeline build time by 30% via no-code tool integration
- Improved Apache Airflow DAG deployment speed by 40% through CI/CD pipeline optimization with GitHub Actions
- Developed a REST API-driven alert system with Cloud Run to notify teams of Adverity DataStream failures, boosting incident response time
- Developed a secure web application for the Data Strategy team, enabling OAuth-based access for users across 5+ agencies

PROJECTS

Finance Tracker | FastAPI, dbt, PostgreSQL, Next.js, Docker | [Link](#)

- Designed a production-style data stack ingesting raw financial data through dbt models into aggregated PostgreSQL marts, served via a JWT-secured FastAPI and visualised in a Next.js dashboard, all containerised with Docker Compose
- Wrote a pytest suite covering auth, token validation, all endpoints, and pagination against a fully mocked database - 13/13 tests passing.

Event-Driven Notification Service | Python, FastAPI, Kafka, aiokafka, PostgreSQL, Docker | [Link](#)

- Built an async event-driven notification service processing order events via Kafka, with idempotent PostgreSQL upserts to safely handle at-least-once delivery and message redelivery
- Implemented retry logic with a dead-letter queue for failed messages, exposing order and notification status through a FastAPI service layer

ACTIVITIES

WPP & Google Cloud Hackathon (3rd Place) | Python, Cloud Run, Vertex AI, Vertex AI Generative Models, Streamlit

Sep 2024

- Designed and deployed a Cloud Run service pulling real-time data from public BigQuery tables to dynamically generate prompts for Vertex AI, rendering AI-generated advertising creatives through a Streamlit frontend, placed 3rd of all competing teams

Global Game Jam Hackathon - ML & Game Developer | Python, OpenCV, C#, Unity

Jan 2024

- Built a Python/OpenCV ML script using camera input to detect laughs, aligned with the hackathon's humour theme
- Contributed Unity/C# game objects and animations to the game prototype

Engineers Without Borders - Project Lead Engineer | MATLAB, SolidWorks

Sep 2020 – May 2021

- Engineered a MATLAB script to optimize thermal efficiency in material selection – increasing solar cooker design efficiency by 20%.
- Led a team of 5 engineers to design a solar cooker prototype with phase-changing materials using SolidWorks

EDUCATION

University of Greenwich

Sep 2020 – May 2023

Bachelor of Engineering, Civil Engineering, 2.1 Upper Second-Class Honours

London, UK

Dissertation (First Class): Assessing the vulnerability of catchments during flash flood events in South England using Machine Learning methods